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Research Article

Sulfo-Fluorous Tagging Strategy for Site-Selective Enzymatic Glycosylation of *para*-Human Milk Oligosaccharides

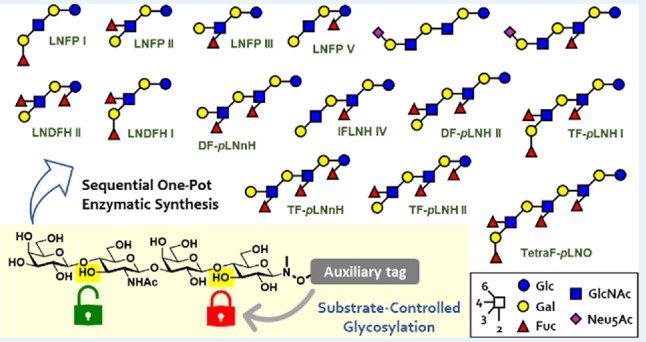
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ABSTRACT: In this study, we developed an efficient fluorous tagging strategy for site-selective enzymatic fucosylation/sialylation in which appending a readily removable auxiliary tag ($\mathbf{SF}_{17}$) to the reducing end of an acceptor provided the regioselectivity control of a given wild-type fucosyltransferase/sialyltransferase. By combining this strategy with the sequential one-pot enzymatic system of HMO production, various fucosylated *para*-HMOs, such as LNFP I, LNFP II, LNFP III, LNFP V, LNDFH I, LNDFH II, TF-LNT, F-*p*LNH I, F-*p*LNH IV, DF-*p*LNH I isomer, DF-*p*LNH II, TF-*p*LNH I, TF-*p*LNH II, F-*p*LNnH, F-*p*LNnH I, DF-*p*LNnH, TF-*p*LNnH, and TetraF-*p*LNO, were synthesized. Furthermore, the $\mathbf{SF}_{17}$-tagged acceptors improved the reaction rates of the enzymatic glycan chain extension. Terminal α2,6-sialylation on $\mathbf{SF}_{17}$-tagged *p*LNnH was directly achieved using wild-type α2,6-sialyltransferase from *Photobacterium damsela* (Pd26ST). Moreover, enzyme kinetics revealed that the catalytic efficiency (k_{cat}/K_M) of α1,4-fucosylation of Bf13FT (α1,3/4-fucosyltransferase from *Bacteroides fragilis*) on $\mathbf{SF}_{17}$-tagged LNFP V acceptors could be enhanced by 55 times compared with tagging with an azidohexyl aglycone. We developed an efficient and reliable strategy for generating complex and diverse natural HMO libraries for functional gut microbiome studies. This $\mathbf{SF}_{17}$-assisted glycosylation strategy provides a practical approach to harness the extreme substrate flexibility offered by readily available bacterial glycosyltransferases, which enables the site-selective fucosylation/sialylation on *para*-HMOs.



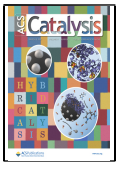
KEYWORDS: *human milk oligosaccharides, enzymatic synthesis, site-selective glycosylation, glycosyltransferases, fluorous solid-phase extraction*

INTRODUCTION

Carbohydrates (glycans) are essential constituents of all living organisms and occur as both simple and complex structures in glycoproteins, glycolipids, and as free glycans. These complex biomolecules play a vital role in numerous physiological processes, such as cellular adhesion, pathogen infection, immune responses, and the development of autoimmune diseases.[1,2] The increased understanding of the biological roles of glycans and their structure−function relationship can be attributed to advances in oligosaccharide synthesis[3] and the development of highly sensitive analytical tools.[4,5] Although chemical synthesis has been a method of exploratory research in this discipline and several types of structurally complicated glycoconjugates can be synthesized,[6−8] chemical glycosylations involve labor-intensive multistep synthetic manipulations and tedious hydroxy group protection and deprotection.[9] The use of carbohydrate-active enzymes, particularly GTs, may circumvent the aforementioned problems due to the strict regioselectivity and stereoselectivity of the enzymes and the mild conditions under which these biomolecules operate, which renders them particularly useful for the formation of glycosides with the desired stereochemistry.[10−13] Furthermore, the enzymatic strategy exhibits high potential for the automated synthesis of oligosaccharides.[14,15]

GTs from mammalian sources exhibit strict substrate specificity that leads to the syntheses of several notable human-type glycan sequences.[16−18] However, enzyme production for industrial-scale synthesis remains a challenging task. By contrast, bacterial GTs are considered particularly promising biocatalysts for facilitating oligosaccharide syntheses because of their broad substrate tolerance, high production levels in *Escherichia coli* overexpression systems, and ease of handling. Several bacterial GTs with promiscuous substrate specificities and multiple functions in forming distinct glycosidic bonds have been extensively used to produce various natural and

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synthetic glycosides.[19−22] A potential limitation of these inherently multifunctional bacterial GTs is the control of a desired catalytic activity on glycan synthesis. Researchers have attempted to tune the catalytic activity of these enzymes through rational protein engineering[23,24] and directed evolution coupled with high-throughput screens.[25] However, the establishment of such approaches requires considerable effort. Substrate engineering has attracted considerable attention because on-demand synthesis using wild-type enzymes can be successfully achieved through simple sugar counterpart modifications.[26] Site-selective enzymatic α2,6-sialylation can be accomplished through redox control,[27] conformation-constrained acceptors,[28] and the installation of protective groups of terminal galactose that resist enzymatic α2,6-sialylation.[29] Similarly, a designed acceptor with protective groups[30] or additional monosaccharide residues[31] may precisely control the positions to be fucosylated under enzyme catalysis, enabling site-selective fucosylated glycan synthesis.

The health benefits of HMOs for breast-fed infants have been investigated. Their role in regulating the intestinal microbiome, modulating epithelial and immune cell responses, and providing nutrients to the brain has been examined.[32−35] All HMO sequences contain lactose at the reducing end. These sequences can be extended through the addition of β1,3- or β1,6-linked lacto-*N*-biose (Galβ1,3-GlcNAc, type-1 LNB) or *N*-acetyllactosamine (Galβ1,4GlcNAc, type-2 LacNAc). Lactose elongation through the β1,3-linkage leads to linear structures, which leads to *para*-HMOs. By contrast, the β1,6-linkage introduces chain branching, which leads to the formation of *iso*-HMOs.[36] Furthermore, the core structure of HMOs can be fucosylated and sialylated, which expands the structural complexity of HMOs (>200 discovered in nature).[37−39] Chemical[40−43] and enzymatic[17,44−48] preparation of HMOs is limited by the difficulty in obtaining long-chain HMOs and their fucosylated/sialylated derivatives. Investigating the roles of long structures of individual HMOs and their potential applications as prebiotics and antimicrobials in nature requires a pure and adequate quantity of structurally defined compounds.[49,50] However, the control of selective incorporation of fucose or sialic acid residues onto the backbone structures by the enzymes is an obstacle for obtaining these long-chain glycan structures. Furthermore, purifying the structurally complex free sugars from a reaction mixture containing sugar nucleotide by-products, cofactors, and buffer salts is difficult.

The discovery of the versatile catalytic properties of FucTs from various bacterial sources will expand the enzymes used for HMO synthesis. We recently reported the synthetic application of α1,2- and α1,3/4-FucTs on fucosylating LacNAc/LNB/lactose-containing acceptors.[51−53] In our study, we focused on acceptor preference profiling of FucTs to predict the products formed on a given oligosaccharide with multiple fucosylation sites. We demonstrated that the addition of α1,2-linked Fuc or β1,3-GlcNAc on the nonreducing end Gal increased the substrate-binding affinity for FucTs, which resulted in a higher turnover number and enhanced catalytic activity. In addition to sugar portion modification, the outcome for a GT-catalyzed reaction could be controlled by reducing end aglycone, where the hydrophobic nature caused various effects.[25,54] Inspired by these unique properties, we devised a facile strategy for site-selective enzymatic glycosylation. This strategy was based on the finding that the *para*-HMO-type glycan with an *N*-methyl oxyamine sulfo-fluorous tag (SF_{17}) installed at the reducing end has a unique substrate characteristic of full or conditional resistance to the enzymatic glycan decoration (α2,6-sialylation or α1,3-fucosylation) on the lactose component, yet allows linear glycan extension (β1,3-*N*-acetyl glucosaminylation followed by β1,3- or β1,4-galactosylation) through corresponding GTs. Therefore, extended type-2 LacNAc and type-1 LNB chains could be modified to fucosides or sialosides with corresponding enzymes (Figure 1). Furthermore, the presence of the SF_{17}

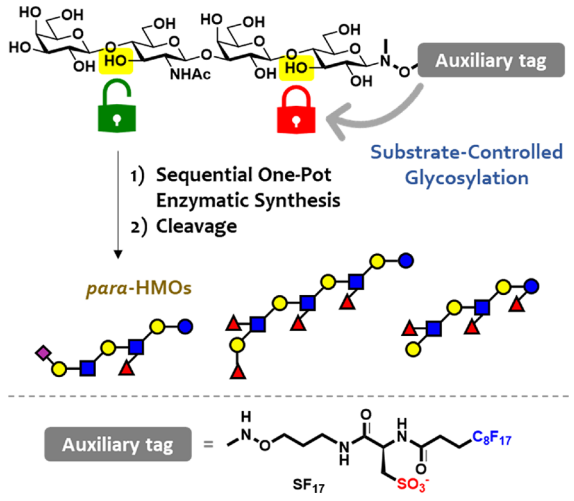


Figure 1. Concept of the SF_{17} tagging strategy for the enzymatic synthesis of *para*-human milk oligosaccharides.

tag enhanced the catalytic efficiency of Bf13FT (an α1,3/4-FucT from *Bacteroides fragilis* NCTC 9343) on type-2 LacNAc over type-1 LNB and lactose components on a given oligosaccharide, which would render site-selective fucosylation possible under controlled conditions. The SF_{17} tag can be easily removed under mild conditions at the end of a sequence of enzymatic glycan extension, which provides complex HMOs with a free reducing end.

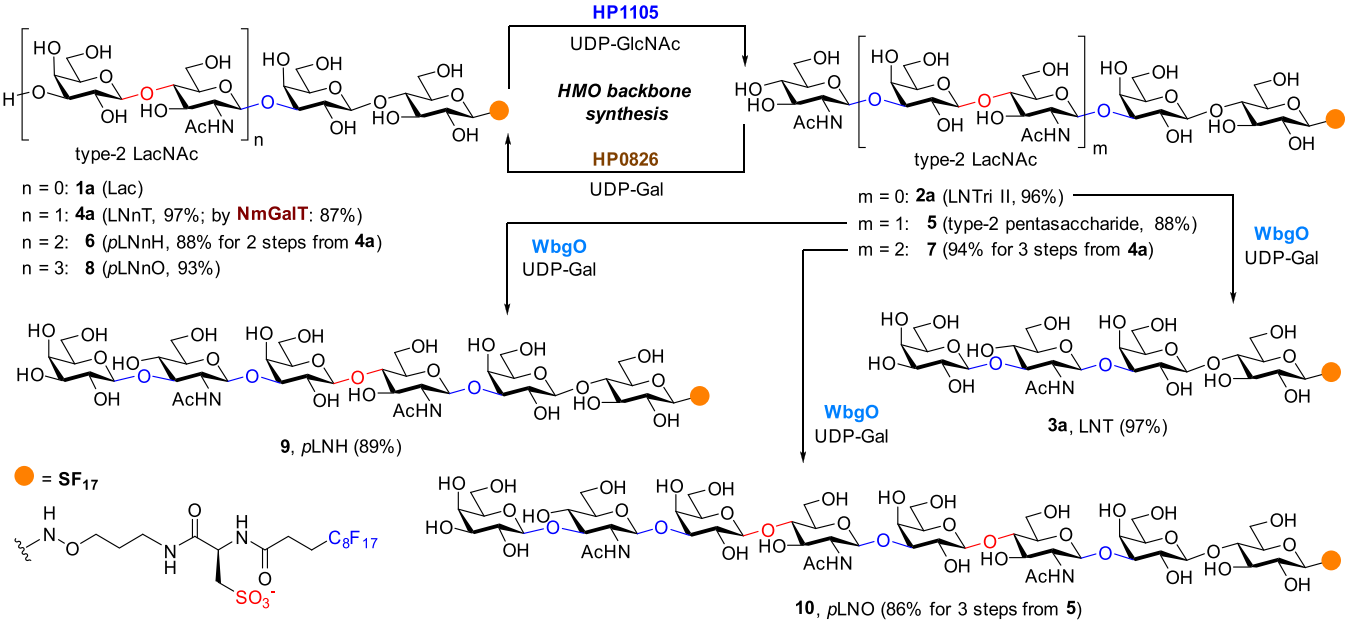
RESULTS AND DISCUSSION

Design of SF_{17} Tag. The unique properties of perfluoroalkyl chains (aka. fluorous tags), which show a high affinity toward the fluorous-phase surface, have made the FSPE an attractive product purification technique.[55,56] Fluorous tag-assisted chemical syntheses are proven to be efficient for the preparation of complex oligosaccharides,[57,58] and these have further led to multistep automated processes.[59,60] The wide applications of the fluorous tag on glycan microarray fabrication,[61] enzyme assay platform,[62] and enzymatic glycan synthesis[63−65] have also been demonstrated. The design of SF_{17} originated from the SOFA tag in which the sulfonate moiety enhanced the water solubility of the fluorous (C_8F_{17})-tagged acceptors.[65] SF_{17} preparation involves combining the sulfonated C_8F_{17} portion of the SOFA tag (sulfo-tag) and the *N*-methyl oxyamine moiety (Figure 1). The resulting SF_{17} that reacts with reducing sugars to form exclusively β-linked cyclic structures[66,67] can be regenerated under controlled reaction conditions.[68,69] The syntheses of the SF_{17} tag and its lactose conjugate for subsequent enzymatic transformation, Lac-SF_{17} (**1a**), were conducted from the readily available starting materials (Supporting Information Scheme S1).

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Scheme 1. Systematic Strategy for the Enzymatic Synthesis of Backbone Structures of Human Milk Glycosides



Reaction conditions.
HP1105: TrisHCl 100 mM, $MgCl_2$ 10 mM, pH 7.5, alkaline phosphatase (AP), 25 °C; WbgO: TrisHCl 50 mM, $MnCl_2$ 30 mM, pH 7.5, AP, 37 °C;
HP0826: HEPES 50 mM, $MnCl_2$ 10 mM, pH 7.2, AP, 37 °C; NmGalT: HEPES 100 mM, $MnCl_2$ 10 mM, pH 7.5, AP, 25 °C

Synthesis of Human Milk Oligosaccharide Backbones. To efficiently construct HMO backbones, three bacterial GTs, specifically β1,3-*N*-acetylglucosaminyltransferase (β1,3-GlcNAcT) from *Helicobacter pylori* ATCC 26695 (HP1105), β1,4-galactosyltransferase (β1,4-GalT) from *H. pylori* NCTC 11637 (HP0826), and β1,3-galactosyltransferase (β1,3-GalT) from *E. coli* O55:H7 (WbgO), were selected as biocatalysts and recombinantly overexpressed in *E. coli*, as previously reported.[53] The sugar donors required for the GTs were generated through the sequential one-pot enzymatic reaction using several sugar-nucleotide-generating enzymes.[51] UDP-GlcNAc was prepared through the action of *N*-acetyl hexosamine kinase from *Bifidobacterium longum* (NahK) and *N*-acetyl glucosamine-1-phosphate uridylyltransferase from *Pasteurella multocida* (PmGlmU), whereas UDP-Gal was obtained using galactokinase from *Meiothermus taiwanensis* sp. nov. WR-220 (MtGalK) and UDP-sugar pyrophosphorylase from *Arabidopsis thaliana* (AtUSP).

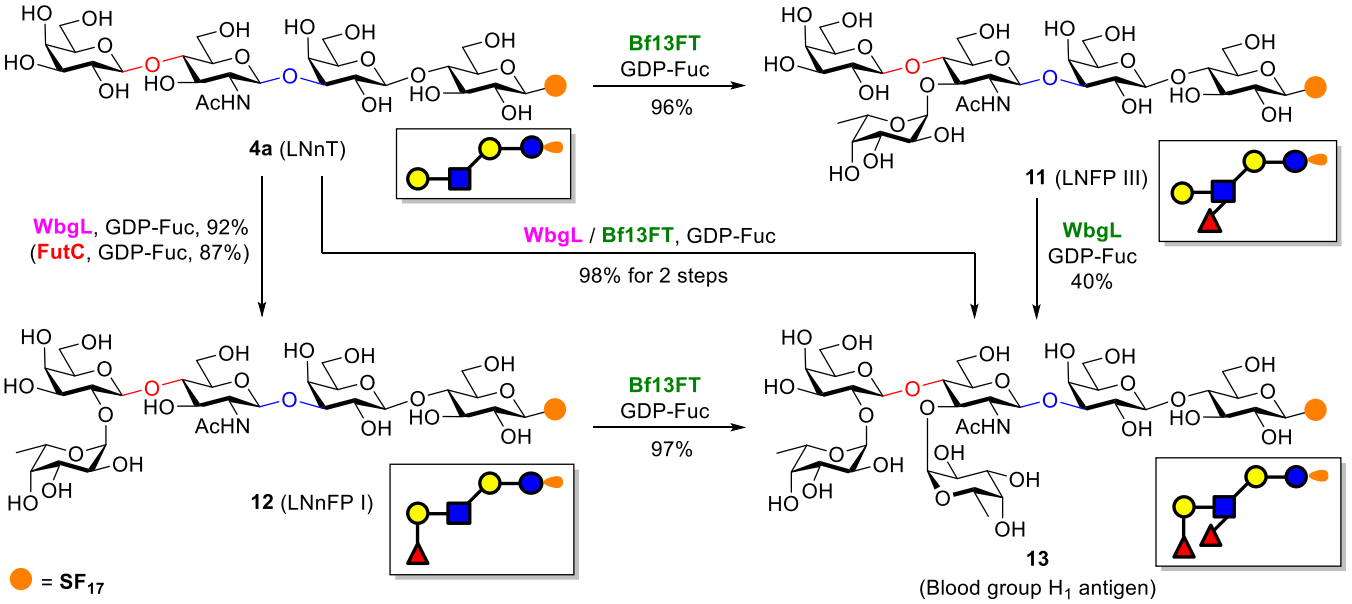
Using HP1105, HP0826, and WbgO with the corresponding UDP-sugars, Lac-SF_{17} (**1a**) was converted into a defined length of the HMO backbone (**2a**, **3a**, **4a**, **5**−**10**), as illustrated in Scheme 1. LNTri II (**2a**) was synthesized through HP1105-catalyzed β1,3-*N*-acetyl glucosaminylation in the presence of UDP-GlcNAc; a 96% yield of the desired trisaccharide was obtained after facile FSPE. LNTri II (**2a**) was subsequently treated with WbgO in the presence of UDP-Gal to obtain a 97% isolated yield of LNT (**3a**). Notably, the catalytic efficiency of HP1105 on the lactose SF_{17} tag was considerably higher than that with an *N*-methyl azidopropyl (Lac-*N*-C_3N_3, **1b**) and an azidohexyl linker (Lac-*O*-C_6N_3, **1c**), as well as that with free lactose. A similar trend was observed on the enzymatic glycosylation catalyzed using WbgO (Supporting Information Figure S1). This trend facilitates HMO synthesis using our strategy because HP1105 and WbgO are key enzymes for HMO backbone construction. Typically, the low conversion rate of WbgO on commonly used substrates (i.e., free sugars or azidohexyl glycans) limits its application for glycan chain extension.[44] Another vital HMO tetrasaccharide, LNnT (**4a**), was synthesized through the catalysis of HP0826 in the presence of UDP-Gal with an isolated yield of 97%. Furthermore, the β1,4-GalT from *N. meningitidis* MC58 (NmGalT) could be used for synthesizing **4a** (87% yield). LNnT **4a** was used as the substrate for HP1105 and reacted with UDP-GlcNAc to obtain an 88% yield of type-2 pentasaccharide (GlcNAcβ1,3-LNnT, **5**). An 88% isolated yield of the hexasaccharide *p*LNnH (**6**) could be prepared from **4a** after two cycles of enzymatic glycosylation (HP1105 followed by HP0826). Similarly, a 93% yield of the octasaccharide *p*LNnO (**8**) was synthesized from **6** through two-step consecutive enzymatic glycosylation. To prepare *p*LNH (**9**), **5** was used in WbgO-catalyzed β1,3-galactosylation and incubated with UDP-Gal to obtain an 89% yield of **9**. To simplify purification, an 86% yield of the *p*LNnO linkage isomer, *p*LNO (**10**), was prepared from **5** through the three-step consecutive enzymatic glycosylation. Structure determination during the syntheses of *p*LNnO (**8**) and *p*LNO (**10**) through NMR was prevented by the severe solubility problems experienced by the end products. We presumed that the gel-like properties of the glycans when dissolving in either deuterated methanol or deuterium oxide resulted from the inherent helical structures of the repeating LacNAc components, which were observed on $(LacNAc)_8$-*O*-C_6N_3, as reported previously.[70]

Site-Selective Fucosylation Strategy Design. To design a systematic synthetic approach for the site-selective fucosylation of HMOs, five bacterial FucTs, namely, Bf13FT,[52] FucTa[71] (also called Hp3/4FT, α1,3/4-FucT from *H. pylori* UA948), FucTIII[51] (α1,3/4-FucT from *H. pylori* DSM 6709), FutC[53] (α1,2-FucT from *H. pylori* 26695), and WbgL[72] (α1,2-FucT from *E. coli* O126) were overexpressed and examined on their substrate specificities using LNT (**3a**) and LNnT (**4a**), as summarized in Figure S2. The recombinant Bf13FT could

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Scheme 2. Systematic Strategy for the Enzymatic Fucosylation on LNnT

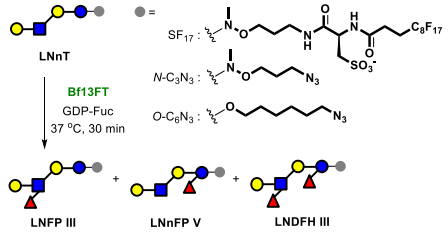


Reaction conditions.
Bf13FT: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 7.5, alkaline phosphatase (AP), 37 °C; FutC: HEPES 50 mM, $MnCl_2$ 5 mM, pH 7, AP, 25 °C; WbgL: TrisHCl 50 mM, $MgCl_2$ 5 mM, pH 7.6, AP, 27 °C

fucosylate the lactose unit of both substrates, the type-1 LNB unit of LNT (**3a**) and the type-2 LacNAc unit of LNnT (**4a**). Among these fucosylated sites to be tested, Bf13FT exhibited excellent catalytic efficiency on type-2 LacNAc compared with LNB and lactose, which allowed for the site-selective fucosylation on the type-2 LacNAc of a given glycan within a short reaction time. Another α1,3/4-FucT, FucTa, fucosylated the type-1 LNB unit of LNT (**3a**) and the type-2 LacNAc unit of LNnT (**4a**) but could not fucosylate the lactose part of the glycans. The activity loss on the lactose unit may have been caused by the presence of SF_{17} at the reducing end, leading to insufficient substrate accommodation in the active site. The other α1,3/4-FucT, FucTIII, exhibited poor catalytic activity on the lactose unit and the type-1 LNB unit of LNT (**3a**). For α1,2-fucosylation, the substrate screening indicated that FutC was capable of fucosylating the terminal galactose of LNT (**3a**) and LNnT (**4a**), but the conversion rate of LNnT (**4a**) was slow. However, WbgL, which prefers type-2 LacNAc units, can be used as a complementary α1,2-FucT for the synthesis of the desired H antigen units. The unique substrate preference profiling of the FucTs on the SF_{17}-tagged glycans allowed us to perform a systematic site-selective fucosylation of *para*-HMOs.

Fucosylation of LNnT. As depicted in Scheme 2, a 96% isolated yield of a monofucosylated product was synthesized from LNnT (**4a**) through the catalysis of Bf13FT in the presence of Tris-HCl (100 mM, pH 7.5), $MgCl_2$ (20 mM), and GDP-Fuc (2.2 equiv) after incubation for 30 min at 37 °C. NMR spectroscopy of the resulting glycan revealed the exclusive formation of α1,3-fucosylation on GlcNAc, which indicated that Bf13FT could selectively α1,3-fucosylate the type-2 LacNAc unit of **4a**, to produce LNFP III (**11**) with excellent efficiency. Extended incubation along with excess GDP-Fuc produce the difucosylated product (data not shown). To emphasize the effect of the aglycone, a truncated version of **4a**, the "sulfo-tag-free" glycoside, LNnT-*N*-C_3N_3 (**4b**), and a commonly used O-linked glycoside, LNnT-*O*-C_6N_3 (**4c**), were examined by comparing the reaction of Bf13FT under the same reaction conditions (Table 1, Figures S3 and S4).

Table 1. Bf13FT-Catalyzed Regioselective α1,3-Fucosylation on LNnT with Different Aglycones



		product yield (%)			
entry	acceptor	LNFP III	LNFP V	LNDFH III	LNnT
1	SF_{17} (**4a**)	96			
2	*N*-C_3N_3 (**4b**)	59[a]	11[a]		29[a]
3	*O*-C_6N_3 (**4c**)	63[b]	11[b]	12[b]	14[b]

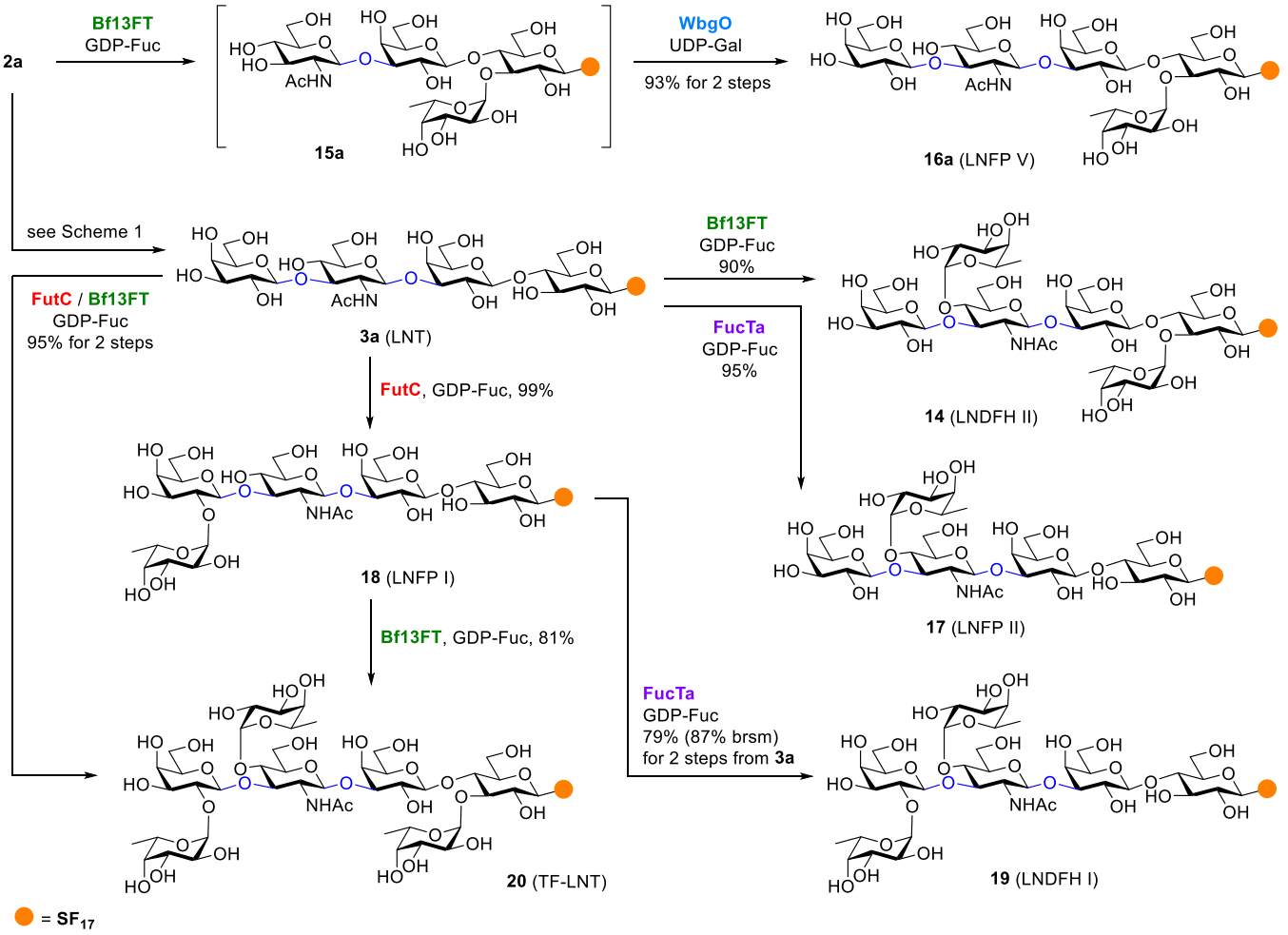
[a]Yield is determined by ^1H NMR. [b]Yield is determined by supercritical fluid chromatography.

Treatment of **4b** with Bf13FT catalysis resulted in two monofucosylated isomers, LNFP III (59%) and LNnFP V (11%), and unreacted LNnT (29%), as displayed in Table 1 (entry 2). The incubation of **4c** with Bf13FT produced two monofucosylated isomers, LNFP III (63%) and LNnFP V (11%), and one difucosylated product, LNDFH III (12%), with 14% of LNnT unreacted (Table 1, entry 3), as determined by supercritical fluid chromatography.[73] These data indicated that the inherent substrate promiscuity of Bf13FT, which led to the synthesis of several fucosylated isomers from LNnT, could be altered by switching the *O*-azidohexyl linker to the SF_{17} tag. This process increased its substrate specificity (Table 1, entry 1). Furthermore,

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Scheme 3. Systematic Strategy for the Enzymatic Fucosylation on LNT



Reaction conditions. WbgO: TrisHCl 50 mM, $MnCl_2$ 30 mM, pH 7.5, alkaline phosphatase (AP), 37 °C; Bf13FT: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 7.5, AP, 37 °C; FucTa: TrisHCl 100 mM, $MnCl_2$ 20 mM, pH 7.5, AP, 37 °C; FutC: HEPES 50 mM, $MnCl_2$ 5 mM, pH 7, AP, 25 °C

modification of the anomeric position with a short *N*-methyl oxyamine linker exhibited no effect on catalytic activity and selectivity enhancement. For enzymes responsible for α1,2-fucosylation, WbgL was capable of synthesizing a 92% yield of LNnFP I (**12**) from LNnT (**4a**) in 2 h; by contrast, the FutC-catalyzed process required 4 days for the reaction and provided an 87% isolated yield. LNnFP I (**12**) was converted to the blood group H_1 antigen (**13**) with a 97% yield through a Bf13FT-catalyzed reaction for 30 min. Notably, alternating the fucosylation sequence (Bf13FT then WbgL) for the synthesis of **13** decreased the overall yield, which resulted from the moderate conversion of WbgL on the Lewis X component of **11** (40% yield). To simplify the synthesis process, a 98% isolated yield of LNnFP I (**12**) could be prepared from **4a** through the one-pot fucosylation by consecutively adding WbgL and Bf13FT.

Fucosylation of LNT. The fucosylation of LNT (**3a**) through the catalysis of Bf13FT with excess GDP-Fuc (4.7 equiv) produced a 90% yield of the difucosylated glycan, LNDFH II (**14**), after 14-h incubation (Scheme 3). Notably, Bf13FT exhibited an enhanced α1,4-fucosylation activity on the type-1 LNB of **3a**, which was reported to be low on the LNT-*O*-C_6N_3 (**3c**) acceptor.[52] The direct conversion of LNT (**3a**) to LNFP V (**16a**) through Bf13FT-catalyzed α1,3-fucosylation would be challenging because the improved α1,4-fucosyltransferase activity on **3a** may result in a mixture of monofucosylated LNT isomers, which would considerably increase the separation efforts. An alternative strategy to obtain LNFP V (**16a**) involved α1,3-fucosylation of LNTri II (**2a**) by Bf13FT, followed by the elongation of the glycan chain of fucosylated LNTri II (**15a**) with a β1,3-linked Gal unit using WbgO, leading to a 93% isolated yield. Furthermore, FucTIII proved to be an efficient enzyme that preferred the transfer of fucose on the Glc unit of LNT and could be used for the synthesis of LNFP V.[51] However, FucTIII exhibited moderate catalytic activity on an SF_{17}-tagged LNT (**3a**), which led to the formation of a 40% yield of LNFP V (**16a**) after incubation for 3 days. FucTa preferentially catalyzed the transfer of fucose to GlcNAc over lactose in LNT; however, longer incubation with a high concentration of GDP-Fuc led to the formation of LNDFH II.[71] In the case of the SF_{17}-tagged substrate, FucTa-catalyzed site-selective fucosylation of LNT (**3a**) was conducted in the presence of GDP-Fuc (2.2 equiv) to obtain a 95% isolated yield of LNFP II (**17**) with no LNDFH II (**14**). This demonstrated that the Glc unit in **3a** is resistant to modifications by FucTa and, therefore, it was possible to selectively α1,4-fucosylate the type-1 LNB unit without considering the concentration of GDP-Fuc used in the reaction. The fucosylation of the terminal galactose in LNT (**3a**) resulted in the formation of a 99% yield of LNFP I (**18**),

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which was catalyzed through the action of FutC. The resulting **18** was further treated with FucTa and Bf13FT to produce LNDFH I (**19**) and trifucosylated LNT (TF-LNT, **20**) with the isolated yields of 79% (87% based on the recovered starting material) and 81%, respectively. Furthermore, a 95% isolated yield of TF-LNT (**20**) could also be prepared through one-pot consecutive fucosylation using FutC and Bf13FT. Compared with the excellent α1,3-fucosylation ability on type-2 LacNAc, Bf13FT required longer incubation (>12 h) to complete the transfer of fucose to the lactose component of LNT. Moreover, the presence of terminal α1,2-linked fucose on LNT considerably reduced the catalysis efficiency, which necessitated a large reaction time (>48 h) to complete the Bf13FT-catalyzed α1,3/4-fucosylaton. A similar trend was observed for FucTa.

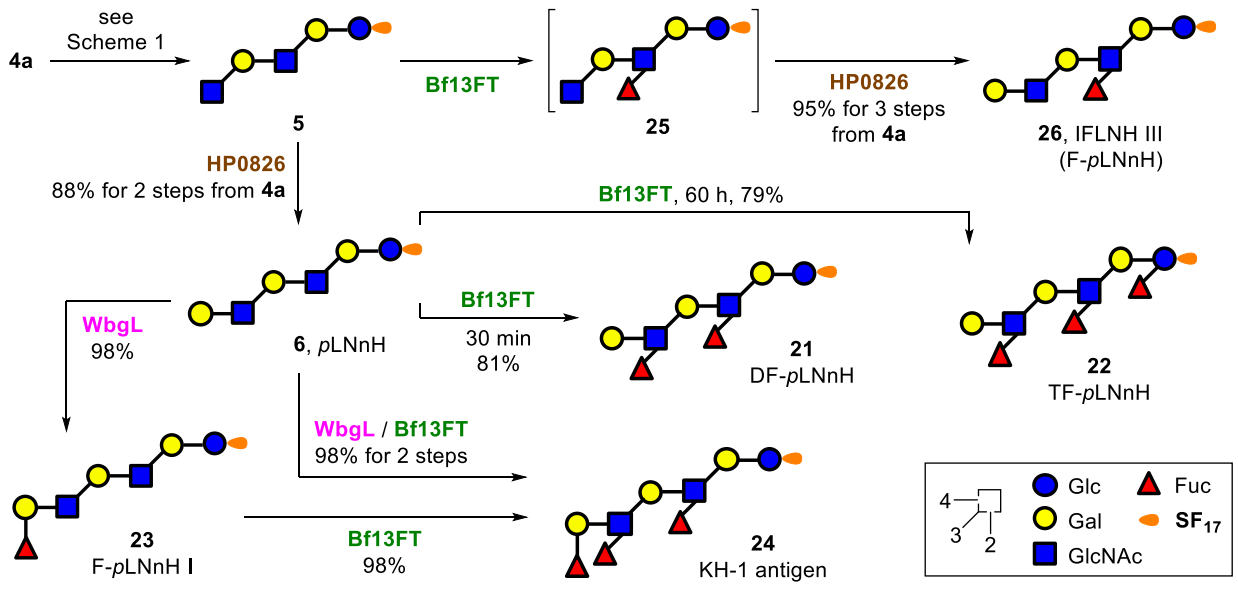
Kinetics of the Bf13FT α1,4-Fucosyltransfer Activity. To understand the enhancement of the α1,4-fucosyltransfer activity of Bf13FT on SF_{17}-tagged glycans, kinetic studies were conducted for three selected sugars, namely, LNFP V (**16a**), LNFP V-*N*-C_3N_3 (**16b**), and LNFP V-*O*-C_6N_3 (**16c**). As summarized in Table 2 (kinetic plots in Figure S5), the azidohexyl-modified LNFP V (**16c**) exhibited a low catalytic efficiency (k_{cat}/K_M = 0.37 mM^{-1} ms^{-1}) with a medium range of K_M (1.77 mM) and a low k_{cat} value (0.65 ms^{-1}). The results clearly indicated a low conversion of LNDFH II during Bf13FT-catalyzed fucosylation when using azidohexyl LNT as the acceptor, which concurred with reports.[52] Notably, replacing the aglycone with an SF_{17} tag (using **16a** as the substrate) increased the overall catalytic efficiency by approximately 55 times compared with the value of k_{cat}/K_M = 20.38 mM^{-1} ms^{-1}. This increase was attributed to the lowest K_M (0.65 mM) and highest k_{cat} (13.36 ms^{-1}). The absence of the sulfo-tag component in the sulfo-tag-free N-linked glycoside **16b** (k_{cat}/K_M = 0.08 mM^{-1} ms^{-1}) reduced the efficiency by 254 times. The considerable decrease in the activity could be attributed to the combination of the poor substrate-binding properties (K_M = 5.02 mM) and decreased turnover number (k_{cat} = 0.42 ms^{-1}). The measurement of the kinetic parameters revealed that the inclusion of SF_{17} not only facilitated control of regiochemistry but also improved reaction rates.

Table 2. Kinetic Parameters of Bf13FT

acceptor	k_{cat} (ms^{-1})	K_M (mM)	k_{cat}/K_M (mM^{-1} ms^{-1})
LNFP V (**16a**)	13.36 ± 0.27	0.65 ± 0.07	20.38 ± 3.70
LNFP V-*N*-C_3N_3 (**16b**)	0.42 ± 0.02	5.02 ± 0.79	0.08 ± 0.02
LNFP V-*O*-C_6N_3 (**16c**)	0.65 ± 0.03	1.77 ± 0.33	0.37 ± 0.08

Fucosylation of *p*LNnH. We then evaluated the potential application of our fluorous tagging strategy on fucosylating two HMO hexasaccharides, *p*LNnH (**6**) and *p*LNH (**9**), as illustrated in Scheme 4. Similar to the synthesis of LNFP III (**11**), Bf13FT preferentially catalyzed fucose transfer to the type-2 LacNAc components of **6**, which led to the formation of DF-*p*LNnH (**21**) in 30 min, with an isolated yield of 81%. Furthermore, the treatment of **6** with BF13FT and an excess amount of GDP-Fuc (5.7 equiv) for 60 h resulted in an isolated yield of 79% in TF-*p*LNnH (**22**). Although the synthesis of IFLNH III (**26**) from **6** by Bf13FT was not feasible due to the indistinguishability of α1,3-fucosylation on either the terminal or internal LacNAc unit on **6**, an alternative approach was achieved by the stepwise introduction of GlcNAc, Fuc (leading to intermediate **25**), and Gal on LNnT (**4a**) through sequential glycosylations involving HP1105, Bf13FT, and HP0826, respectively; a 95% isolated yield of the desired glycan **26** was obtained. Following the procedure for the synthesis of the blood group H_1 antigen (**13**), *p*LNnH (**6**) was α1,2-fucosylated by WbgL to produce a 98% yield of F-*p*LNnH I (**23**). Subsequently, the catalysis of

Scheme 4. Systematic Strategy for the Enzymatic Fucosylation on *p*LNnH



Reaction conditions. HP0826: HEPES 50 mM, $MnCl_2$ 10 mM, pH 7.2, alkaline phosphatase (AP), UDP-Gal, 37 °C; Bf13FT: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 7.5, AP, GDP-Fuc, 37 °C; WbgL: TrisHCl 50 mM, $MgCl_2$ 5 mM, pH 7.6, AP, GDP-Fuc, 27 °C

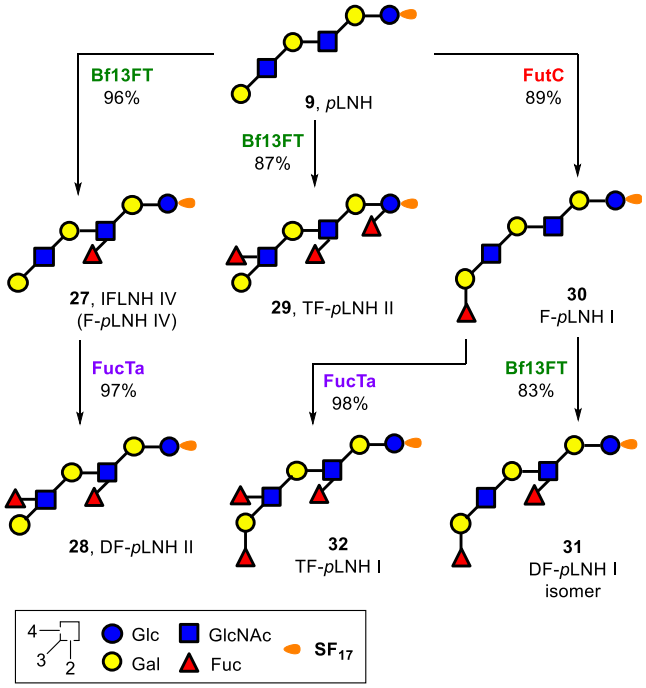
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Bf13FT on F-*p*LNnH I (**23**) resulted in a 98% yield of a critical tumor-associated carbohydrate antigen, KH-1 antigen (**24**). Similarly, an isolated yield of 98% KH-1 antigen could be prepared through the programmable one-pot fucosylation that involved the addition of WbgL followed by Bf13FT. Studies on the enzymatic KH-1 antigen synthesis using F-*p*LNnH I-*O*-C_6N_3 as a fucosylation acceptor have obtained 24−34% yields of three multifucosylated by-products in addition to 59−66% yields of the desired KH-1.[52] This promiscuous product distribution using the F-*p*LNnH I-*O*-C_6N_3 acceptor accentuated the advantage of using SF_{17}-tagged F-*p*LNnH I to change the inherent substrate specificity of Bf13FT.

Fucosylation of *p*LNH. Following the fucosylation of LNT, *p*LNH (**9**) could be converted into six vital fucosylated HMOs, **27**−**32**, using Bf13FT, FucTa, and FutC systematically, as illustrated in Scheme 5. Briefly, monofucosylation and

Scheme 5. Systematic Strategy for the Enzymatic Fucosylation on *p*LNH



Reaction conditions. **Bf13FT**: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 7.5, alkaline phosphatase (AP), GDP-Fuc, 37 °C; **FucTa**: TrisHCl 100 mM, $MnCl_2$ 20 mM, pH 7.5, AP, GDP-Fuc, 37 °C; **FutC**: HEPES 50 mM, $MnCl_2$ 5 mM, pH 7, AP, GDP-Fuc, 25 °C

trifucosylation of **9** through the catalysis of Bf13FT under controlled α1,3/4-fucosylation resulted in isolated yields of 96 and 87% of IFLNH IV (**27**) and TF-*p*LNH II (**29**), respectively. Subsequent α1,4-fucosylation on **27** using FucTa produced a 97% isolated yield of difucosylation product DF-*p*LNH II (**28**). An 89% isolated yield of F-*p*LNH I (**30**) was synthesized through FutC-catalyzed α1,2-fucosylation. Furthermore, the selective transfer of fucose to the internal type-2 LacNAc unit of **30** could be achieved through the action of Bf13FT, which led to the formation of an 83% yield of the DF-*p*LNH I isomer (**31**). Accordingly, the incubation of **30** through the FucTa-catalyzed α1,3/4-fucosylation produced a 98% isolated yield of trifucosylated glycan, TF-*p*LNH I (**32**).

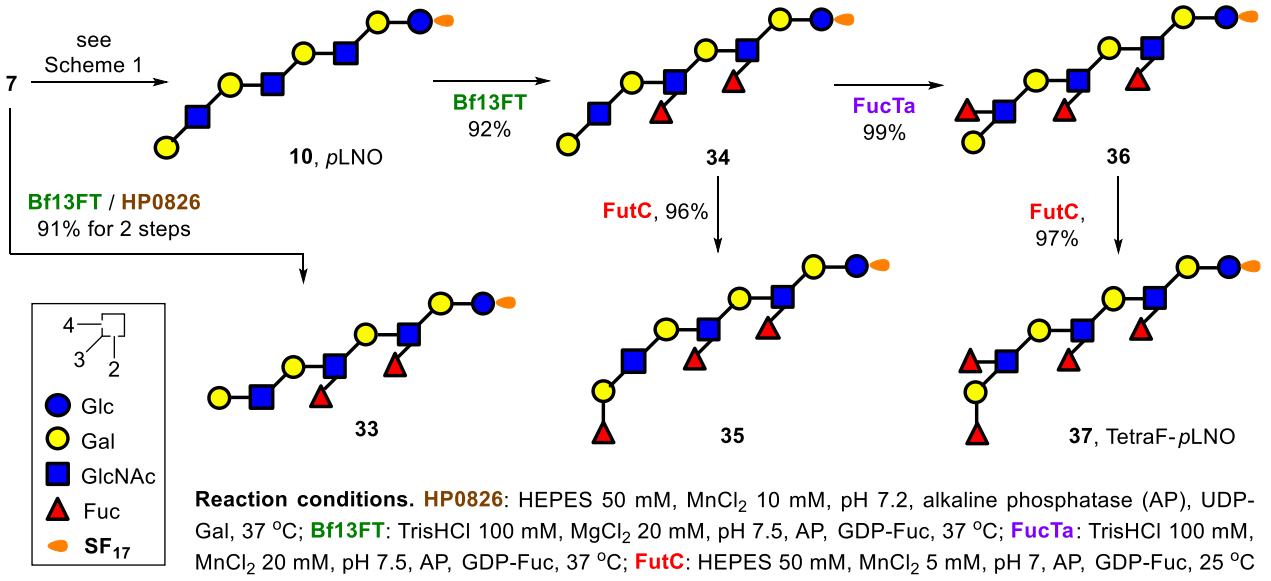
Fucosylation on Octasaccharides. Linear octasaccharide core structures and their fucosylated derivatives have been identified in human milk. However, structural investigations and in vitro preparations have been rarely reported. As displayed in Scheme 6, the preparation of internal α1,3-difucosylated *p*LNnO (**33**) was initiated by Bf13FT-catalyzed fucosylation on heptasaccharide **7**, followed by efficient HP0826-catalyzed β1,4-galactosylation, which produced a yield of 91% glycan (**33**). Notably, the decoration of fucose on *p*LNnO increased its solubility, which concurred with the reported literature.[74] Presumably, the additional fucose residues slightly distorted the helix structure formed by oligo-LacNAc repeating units, which led to superior solubility. NMR spectroscopy revealed that TetraF-*p*LNO composed of a *p*LNO backbone attached to four fucose components (III^3Fuc, V^3Fuc, VII^4Fuc, and VIII^2Fuc) with an α-linked glycosidic bond is the most complex linear fucosylated HMOs to date.[75] We synthesized TetraF-*p*LNO using our fluorous tagging strategy that involved a sequential α1,3-, α1,4-, and α1,2-fucosylation on *p*LNO glycans. Here, α1,3-fucosylation of *p*LNO (**10**) was catalyzed by Bf13FT and provided a 92% yield of internal α1,3-di-fucosylated *p*LNO (**34**). The resulting glycan could be either α1,2-fucosylated by FutC to produce a 96% yield of trifucosylated *p*LNO (**35**) or α1,4-fucosylated by FucTa to provide an excellent 99% yield of trifucosylated *p*LNO (**36**). Further decoration of α1,4-fucosylated by FucTa with a terminal α1,2-linked fucose on the Gal under the catalysis of FutC produced an isolated 97% yield of TetraF-*p*LNO (**37**).

Terminal Sialylation of *p*LNnH. The structure of *p*LNnH with a terminal α2,6-sialyl residue (namely 4021b) has been isolated in human milk,[76] but its biological function remains unknown. Notably, the mammalian SiaT, hST6Gal-I, could be used to perform the terminal sialylation.[77] However, this selectivity is considered a difficult task for bacterial enzymes. One of the commonly used bacterial α2,6-SiaT, Pd26ST, has been used for several applications such as sialylating a library of glycans because of its excellent catalytic efficiency and broad substrate tolerance.[21] However, the tradeoff, in this case, is the regioselectivity upon sialylation. In the preliminary tests of sialylating SF_{17}-tagged glycans, we determined that Lac-SF_{17} (**1a**) was resistant to α2,6-sialylation by Pd26ST, whereas other substrates such as **1b**, **1c**, and free lactose could be converted to the corresponding α2,6-sialylated lactosides (Figure S6). We presumed that the low Pd26ST activity that resulted from the improper orientation of **1a** in the active site could be restored as the glycan chain extended. Notably, instead of forming the disialylated glycan as expected, the sialylated product was not detected during the catalysis of Pd26ST using pentasaccharide (**5**) as the acceptor, as illustrated in Scheme 7. Furthermore, the addition of Gal introduced by HP0826 could trigger subsequent Pd26ST catalysis, which could facilitate selective α2,6-sialylation of the terminal Gal of *p*LNnH (**6**). This produced an isolated 94% yield of monosialylated *p*LNnH (**38**) as the only product (for two steps) with three equivalents of CMP-Neu5Ac. The internal type-2 LacNAc component in **38** could be converted to a Lewis X component using Bf13FT to obtain a 94% yield of monosialylated IFLNH III (**39**). Furthermore, based on NMR characterization, the successful transfer of the internal fucose residue to **38** also demonstrated that the terminal sialic acid was attached to *p*LNnH **6** upon Pd26ST catalysis, which resulted in an internal LacNAc unit for further modification. Furthermore, a 90% isolated yield of Sia-IFLNH III **39** could be obtained from **26** through α2,6-sialylation by Pd26ST. The

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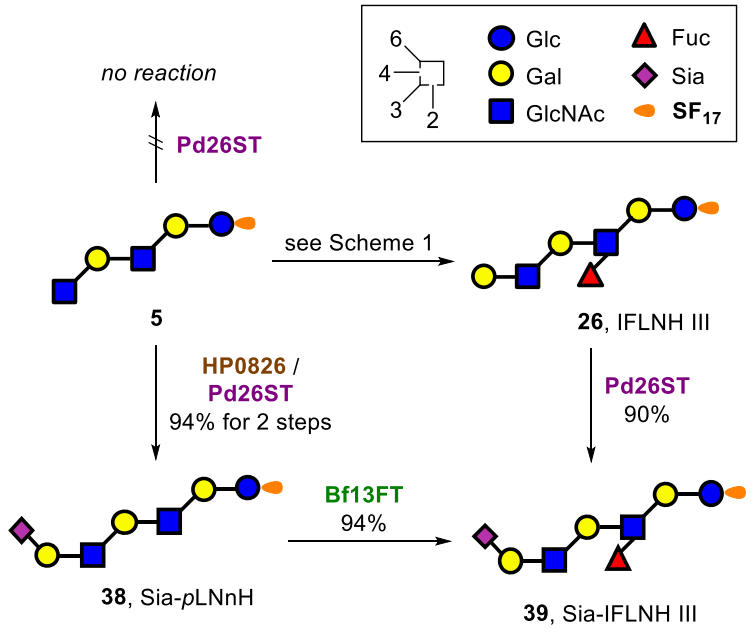
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Scheme 6. Systematic Strategy for the Enzymatic Fucosylation on HMO Octasaccharides



Reaction conditions. HP0826: HEPES 50 mM, $MnCl_2$ 10 mM, pH 7.2, alkaline phosphatase (AP), UDP-Gal, 37 °C; Bf13FT: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 7.5, AP, GDP-Fuc, 37 °C; FucTa: TrisHCl 100 mM, $MnCl_2$ 20 mM, pH 7.5, AP, GDP-Fuc, 37 °C; FutC: HEPES 50 mM, $MnCl_2$ 5 mM, pH 7, AP, GDP-Fuc, 25 °C

Scheme 7. Systematic Strategy for the Enzymatic Synthesis of Sialylated Human Milk Glycosides



Reaction conditions. HP0826: HEPES 50 mM, $MnCl_2$ 10 mM, pH 7.2, alkaline phosphatase (AP), UDP-Gal, 37 °C; Bf13FT: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 7.5, AP, GDP-Fuc, 37 °C; Pd26ST: TrisHCl 100 mM, $MgCl_2$ 20 mM, pH 8.0, AP, CMP-Neu5Ac, 37 °C

results indicated that hexasaccharide was required for Pd26ST to perform α2,6-sialylation on the SF_{17}-tagged glycan. This approach is the simplest method to achieve enzymatic α2,6-sialylation on the terminal Gal of *p*LNnH without blocking the internal Gal positions (^{II}Gal and ^{IV}Gal) by incorporating protective group manipulations. Docking experiments revealed that the hydrophobic regions near the Pd26ST active site may have considerable interactions with the SF_{17} tag, leading to various orientations of the tagged glycan within the active site, unlike that with reducing sugar (Figure S8). The detailed procedures and parameters used in docking are described in the Supporting Information (Figures S7 and S8). The products were analyzed in terms of their purity and structure through ESI-MS and NMR spectroscopy. The structures of the compounds were confirmed through one- and two-dimensional NMR analyses (Tables S2−S6).

Natural HMO Production via Removal of SF_{17} Tag. The collection of structurally defined HMOs for use as standards to evaluate their vital roles in the living system through major biological assays, such as those testing the

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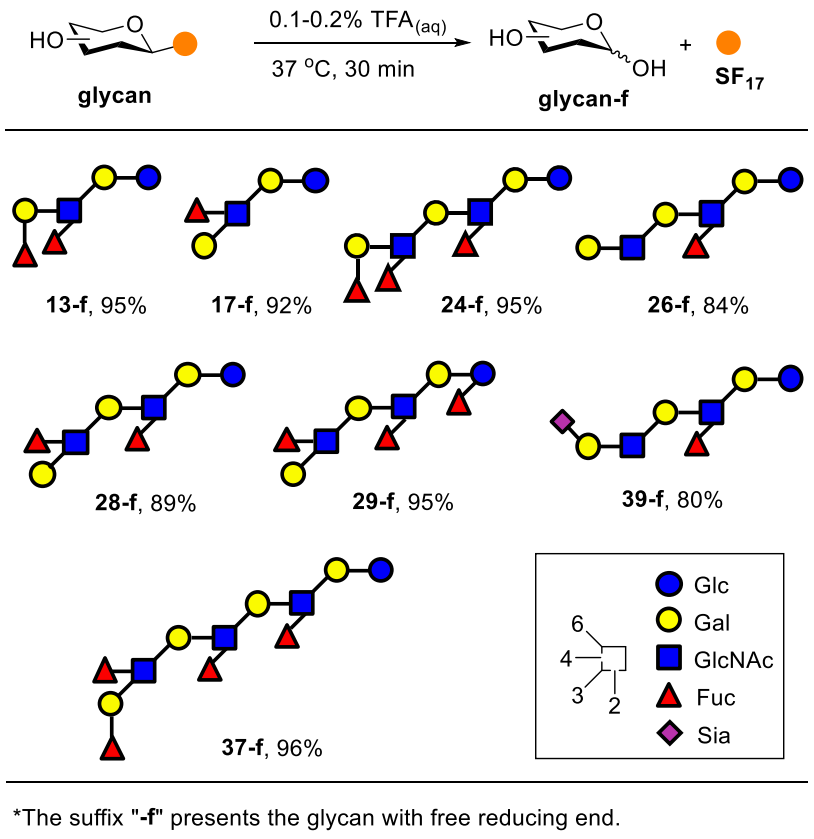


Figure 2. Acidic cleavage of SF_{17}-tagged glycans.

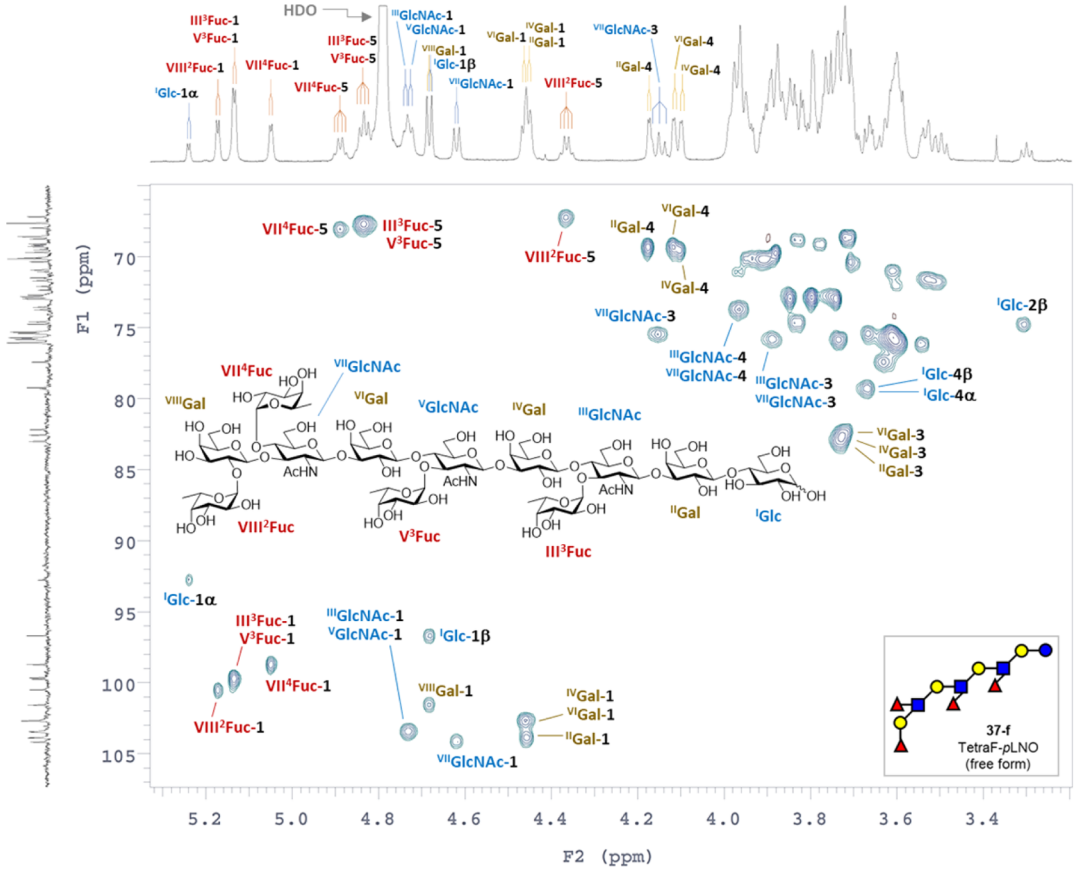


Figure 3. Structure of tetrafucosyl-*para*-lacto-*N*-octaose **37-f** and its HSQC spectrum.

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prebiotic activity and mechanism of oligosaccharide digestion by intestinal bacteria, are exceedingly important. Various SF_{17}-tagged HMOs can be synthesized using our strategy. We therefore focused on cleaving off the glycans from the tag. These free reducing-end glycans can be applied in future biological studies. We therefore conducted ESI-MS structural analysis on 22 reducing glycans derived from the anomeric tag removal experiments (Table S1). Briefly, the SF_{17} tags could be simply removed through treatment with 0.1−0.2% trifluoroacetic acid in water for 30 min, with a yield of 71−99% glycans with a free reducing end (**glycan-f**). Furthermore, the released SF_{17} could be recycled after FSPE purification (51−97% yields) and reused for the glycosylation of the next round. Eight of the free-form glycans were characterized through NMR for structure elucidation (Figures 2 and 3, Tables S7−S15). The data revealed that the cleavage conditions did not affect the fucosidic bonds within the corresponding acid-labile glycans. Notably, the acidic cleavage of monosialylated IFLNH III (**39**) produced the tag and desired free sugars **39-f** (80%), along with the free form 18% yield of IFLNH III (**26-f**). This result implied that the fucosidic bond was more stable than the sialosidic bond in an acidic environment.

CONCLUSIONS

In conclusion, we developed an SF_{17} tagging strategy for enzymatic glycan syntheses, which facilitated the rapid preparation of 38 linear HMO glycosides. Conventional methods for modulating the substrate preference of GTs for on-demand synthesis required changes in both the substrate and the catalyst. By contrast, by simply attaching a readily removable SF_{17} tag to the reducing end of the lactose-containing glycan acceptors, we were able to alter the substrate specificity of wild-type Bf13FT and FucTa, which led to site-selective incorporation of fucose moieties onto the *para*-HMO core structures under controlled conditions. Terminal α2,6-sialylation on SF_{17}-tagged *p*LNnH could be achieved directly using the wild-type Pd26ST, which resulted in multisialylated products when the commonly used alkyl chain aglycone was applied. Notably, SF_{17}-tagged acceptors not only improved the rates of the WbgO- and HP1105-catalyzed reactions but also enhanced the inherent α1,4-fucosyltransfer activity of Bf13FT substantially. We reasoned that the increased catalytic activity of the GTs with SF_{17} tagged substrates resulted from the detergent-like properties of the SF_{17}, which stabilized these membrane-associated enzymes by providing a membrane-like environment.[78] Additionally, the hydrophobic portion of the SF_{17} not only increased the affinity but also altered the orientation of the tagged glycans toward the enzyme active sites upon catalysis.

These results demonstrated the practical application of the SF_{17} tagging strategy to enzymatic glycan synthesis and highlighted that various complex *para*-HMOs can be prepared efficiently, yielding high-purity products. Although a number of tagged and free reducing *para*-HMOs were synthesized, methods for preparing complex architectures with α1,3- and α1,4-fucosylation have been rarely described. Our strategy addresses these deficiencies and provides a method for preparing diverse fucosyl HMOs. A list of *para*-HMOs synthesized by known enzymatic methods is included in Table S16 for comparison. Having these well-defined structures provides valuable standards for the characterization and quantification of complex glycans isolated from nature and enables comprehensive screens for new prebiotic activities and advances in exploring the HMO−gut microbiome relationship. These can considerably expedite the progress of novel formula production in the future.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.0c04934.

Detailed experimental procedures, enzyme purification, enzyme kinetics plots, and ^{1}H NMR and ^{13}C NMR spectra illustrating chemical structures for enzymatic products (PDF)

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Author Contributions

Y.-T.H. performed synthesis, protein expression as well as kinetic experiments. Y.-C.S. and H.-H.H. performed gene cloning, assisted with protein expression, and synthesized donor compounds. H.-R.W. performed and analyzed ESI-MS experiments. E.C.L. performed docking experiments with enzyme substrates. T.-W.T., H.-W.T., and J.-L.F. synthesized donor compounds. Y.-T.H. and C.-C.Y. designed the synthesis, analyzed the resulting data, and wrote the manuscript.

Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

HMO, human milk oligosaccharide; GlcNAc, *N*-acetylglucosamine; Gal, galactose; GT, glycosyltransferase; FucT, fucosyltransferase; SiaT, sialyltransferase; FSPE, fluorous solid-phase extraction; SOFA, sulfo-fluorous affinity; LNTri II, lacto-*N*-triose II; LNT, lacto-*N*-tetraose; LNnT, lacto-*N*-neotetraose; *p*LNnH, *para*-lacto-*N*-neohexaose; *p*LNnH, *para*-lacto-*N*-neohexaose; *p*LNO, *para*-lacto-*N*-octaose; *p*LNnO, *para*-lacto-*N*-neooctaose; LNFP, lacto-*N*-fucopentose; LNnFP, lacto-*N*-neopentose; LNDFH, lacto-*N*-difucohexaose; F-*p*LNH, fucosyl *para*-lacto-*N*-hexaose; IFLNH III, isofucosyl-lacto-*N*-hexaose III; DF-*p*LNnH, difucosyl-*para*-lacto-*N*-neohexaose; TF-*p*LNnH, trifucosyl-*para*-lacto-*N*-neohexaose; TetraF-*p*LNO, tetrafucosyl-*para*-lacto-*N*-octaose; CMP-Neu5Ac, cytidine 5′-monophosphate-*N*-acetyl-neuraminic acid; GDP-Fuc, guanosine diphosphate fucose; UDP-GlcNAc, uridine diphosphate *N*-acetyl-glucosamine; NMR, nuclear magnetic resonance; ESI-MS, electrospray ionization mass spectroscopy

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